

HALLHAXASEN, Sweden

WMO: 022360

Lat: 63.7685N

Lon: 15.3279E

Elev: 1234

StdP: 14.05

Time Zone: 1.00 (EUC)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -9.9	(c) -4.8	(d) -14.6	(e) 2.6	(f) -9.8	(g) -9.2	(h) 3.5	(i) -4.2	(j) 18.3	(k) 33.1	(l) 15.3	(m) 31.4	(n) 2.2	(o) 40	(p) 0.655

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 16.4	(c) 76.3	(d) 61.0	(e) 72.8	(f) 59.7	(g) 69.1	(h) 58.0	(i) 63.5	(j) 72.2	(k) 61.5	(l) 69.1	(m) 59.6	(n) 66.7	(o) 5.5	(p) 170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
60.1	81.3	66.4	58.2	75.9	64.5	56.4	70.9	62.7	29.4	72.3	28.0	68.7	26.7	66.6	69.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 15.3	(b) 13.0	(c) 11.3	(e) -16.1	(f) 80.4	(g) 6.0	(h) 3.6	(i) -20.4	(j) 82.9	(k) -23.9	(l) 85.0	(m) -27.3	(n) 87.0	(o) -31.6	(p) 89.6		
(4) 15.3	(5) 13.0	(6) 11.3	(7) -16.4	(8) 66.4	(9) 5.7	(10) 2.5	(11) -20.5	(12) 68.1	(13) -23.9	(14) 69.6	(15) -27.1	(16) 70.9	(17) -31.2	(18) 72.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	36.7	17.6	19.1	25.7	35.7	45.2	52.7	57.9	55.2	46.6	35.7	26.6	20.9
	DBStd	16.20	11.23	11.08	8.45	6.21	7.31	6.66	6.01	5.73	5.40	6.69	8.08	10.99
	HDD50	5479	1003	864	754	430	189	44	5	15	127	443	702	903
	HDD65	10356	1468	1284	1219	879	612	374	233	309	551	907	1152	1368
	CDD50	617	0	0	0	0	41	124	250	175	26	1	0	0
	CDD65	21	0	0	0	0	0	4	13	4	0	0	0	0
	CDH74	199	0	0	0	0	6	46	119	28	0	0	0	0
	CDH80	14	0	0	0	0	0	0	13	1	0	0	0	0
Wind	WSAvg	4.9	4.3	4.8	5.9	5.5	5.7	5.6	4.6	4.4	4.8	5.0	4.3	4.3
Precipitation	PrecAvg	23.2	1.4	1.1	1.1	1.3	1.9	2.4	3.4	2.6	2.4	2.0	1.9	1.5
	PrecMax	31.1	2.7	1.9	1.7	2.3	3.6	3.9	7.4	5.7	5.4	7.3	3.8	3.5
	PrecMin	15.9	0.3	0.3	0.3	0.4	0.6	0.4	1.1	0.2	0.6	0.0	0.9	0.4
	PrecStd	4.3	0.6	0.5	0.4	0.5	0.9	0.9	1.9	1.7	1.2	1.5	0.9	0.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	40.7	41.6	46.5	62.1	74.4	78.5	82.0	77.6	65.7	56.2	47.7	41.2
		MCWB	37.4	38.0	38.9	48.1	57.0	59.6	65.4	62.8	57.0	50.1	45.9	38.9
	2%	DB	37.5	39.2	42.3	55.4	69.0	74.6	77.3	73.4	61.4	50.4	42.3	38.4
		MCWB	35.5	36.1	37.4	44.3	53.6	58.2	62.5	61.2	54.8	47.0	41.2	36.3
	5%	DB	34.6	36.5	39.8	50.5	64.1	69.5	73.6	69.3	58.2	47.9	38.9	36.3
		MCWB	33.0	34.3	35.8	41.5	51.5	56.0	61.0	59.5	53.2	45.3	37.7	34.8
	10%	DB	31.9	33.2	37.6	46.4	58.7	65.3	69.7	65.6	55.4	45.7	36.4	33.9
		MCWB	30.9	31.5	34.1	39.5	49.3	54.4	59.6	58.1	51.3	43.5	35.5	32.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	37.7	38.5	40.4	48.4	58.4	62.9	66.9	65.3	58.8	51.0	45.7	39.0
		MCDB	40.0	40.8	44.8	59.3	69.1	73.7	77.7	72.7	63.2	53.9	47.3	40.7
	2%	WB	35.7	36.5	38.1	45.4	55.8	59.8	64.2	62.8	56.0	47.8	41.4	36.6
		MCDB	37.3	38.8	41.6	54.3	65.5	69.7	74.1	69.8	59.9	49.8	42.3	38.0
	5%	WB	33.2	34.4	36.3	42.6	52.9	57.6	62.3	60.8	53.9	45.8	38.1	34.9
		MCDB	34.4	36.3	39.4	49.0	62.3	67.3	70.8	67.2	57.4	47.6	38.7	36.1
	10%	WB	30.9	31.6	34.3	40.0	50.1	55.4	60.5	58.9	51.9	43.6	35.6	32.8
		MCDB	31.9	33.1	37.3	45.3	57.9	64.1	67.9	64.5	54.7	45.6	36.4	33.7
Mean Daily Temperature Range	5% DB	MDBR	10.2	11.0	13.8	14.4	16.7	16.9	16.4	15.6	13.1	10.2	7.9	9.6
		MCDBR	9.5	10.5	13.8	22.1	23.9	23.1	21.9	20.5	17.0	13.2	9.3	9.7
		MCWBR	8.7	9.0	10.2	12.6	12.4	11.0	10.3	10.2	11.4	10.1	8.6	9.0
	5% WB	MCDBR	9.6	10.2	13.2	19.6	21.9	20.4	19.4	17.4	15.7	12.2	9.1	9.6
MCWBR		8.9	9.0	10.1	12.1	12.4	10.7	10.0	9.6	11.2	9.9	8.6	8.9	
Clear-Sky Solar Irradiance	taub	0.199	0.264	0.292	0.326	0.331	0.328	0.341	0.330	0.303	0.287	0.354	0.349	
	taud	2.017	2.230	2.312	2.401	2.483	2.517	2.493	2.513	2.576	2.473	2.817	2.775	
	Ebn at Noon	131	213	256	272	281	284	276	269	256	209	126	75	
	Edh at Noon	9	18	25	29	30	29	29	26	20	15	8	5	
All-Sky Solar Radiation	RadAvg	49	226	651	1193	1478	1674	1554	1156	672	283	75	22	
	RadStd	5	31	69	102	138	121	169	100	71	41	11	1	

Historical Trends

		Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (48 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	-247	N/A	N/A	N/A

Nomenclature: See separate page